

Live Architecture, Agents, Automation, and Gaps

Prepared from direct VPS inspection of /var/www/khojai. Version 1.1, 31 May 2026.

Executive Summary

KhojAI is already a live VPS project. It runs a public Urdu Next.js website, an AI content pipeline, generated lesson/image outputs, a scheduler, SQLite database, and WhatsApp teacher code.

Current public site: <https://khojai.zuhaib.pro/>. Desired consolidated product destination: <https://khoj.zuhaib.pro/>. Internal control tower: <https://lab.zuhaib.pro/khoj>.

Actual VPS Folder Map

/var/www/khojai/	
-- app/	Next.js Urdu website: home, about, lessons, sitemap
-- agents/	director, curriculum, content, visual, social, SEO, QA, security, notify
-- aiteacher/	Baileys WhatsApp teacher and auth/session directory
-- bot/	Legacy Meta Cloud API webhook bot
-- components/	Present, not deeply inspected
-- content/output/	Generated lesson JSON and slide PNG images
-- lib/	Present, not deeply inspected
-- logs/	director, scheduler, pipeline, errors logs
-- public/	Logo and static assets
-- pipeline.js	Older daily pipeline entrypoint
-- scheduler.js	Current every-2-days/support scheduler
-- khoj.db	SQLite database
-- .env	Runtime secrets, not opened
-- next.config.mjs	Next config
-- package.json	Node dependencies and scripts

Runtime Status

Process	Status	Purpose	Observed Detail
khojai-web	online	Public Next.js website	npm start from /var/www/khojai on port 3001.
khojai-agents	online	Scheduler and agent runner	Runs /var/www/khojai/scheduler.js.
khojai-aiteacher	stopped	WhatsApp Baileys teacher	PM2 process exists but is not online. Logs not read to avoid private WhatsApp exposure.
khoj	online	Internal dashboard	Next app on port 3002, proxied at lab.zuhaib.pro/khoj.
khoj-executor	online	Product/executor service	Node service on port 3010, proxied at khoj.zuhaib.pro.

Nginx Routing

Domain	Route	Backend	Notes
khojai.zuhaib.pro	/	localhost:3001	Public Urdu Next.js website.
khojai.zuhaib.pro	/content/	/var/www/khojai/content/output/	Static generated output with cache headers.
khojai.zuhaib.pro	/webhook, /bot/	localhost:3003	Legacy Meta bot route; port 3003 was not listening.

Domain	Route	Backend	Notes
khaj.zuhaib.pro	/	127.0.0.1:3010	Current product/executor service.
lab.zuhaib.pro	/khaj, /_next/	127.0.0.1:3002	Internal tracking dashboard.

Agent Council

Agent	File	Role
Director	agents/director.js	Runs curriculum, content, QA, visual, social, DB save, website rebuild, founder notify.
Curriculum	agents/curriculum.js	Selects one NANO topic in Urdu using Claude.
Content	agents/content.js	Writes 9-slide Urdu carousel, caption, hashtags, WhatsApp message.
QA	agents/qa.js	Checks Urdu simplicity, emoji rule, science accuracy, safety, score.
Visual	agents/visual.js	Generates slide PNGs with OpenAI gpt-image-1.
Social	agents/social-media.js	Creates Urdu posting strategy and hashtags.
SEO	agents/seo.js	Generates trends and stores seo_data.
Security	agents/security.js	Checks env, SSL, disk, PM2; disk check command needs fixing.
Notify	agents/notify.js	Sends founder WhatsApp updates; recipient should move to env.

Automation Schedule

Runner	Schedule	Action
scheduler.js	0 0 */2 **	runContentPipeline() every two days at midnight.
scheduler.js	0 3 * * 1,4	runSupportAgents() Monday and Thursday at 3 AM.
pipeline.js	0 0 * * *	Older daily pipeline if the old process is started.

Database Schema

Table	Purpose
content	Lesson date, topic, slides JSON, image URLs, Instagram caption, WhatsApp message, status, created time.
students	Legacy student profile table.
conversations	Legacy Meta bot conversation history.
wa_users	Baileys teacher users: phone, name, relation, first message, message count.
wa_conversations	Baileys teacher conversation history.
seo_data	Created by SEO agent after successful support run; not present in schema output during audit.

Generated Content Observed

VPS contains generated output for 2026-05-31 plus PNG slide images in content/output. Inspected lesson topic: ????? ?? ?????? ???. Logs also show a later Director run for ?????? ??? ?? ?????? ????? with 9/9 images and QA score 9/10.

Signal	Finding
Director log	Pipeline completed: topic, content, QA, 9/9 images, social strategy, website rebuild.
Pipeline log	Older pipeline had overlapping manual runs; one run generated 0/9 images, another 9/9.
Error log	One JSON parse error from markdown fenced output.
Quality gap	Latest output includes emojis and English hashtags despite no-emoji and Urdu-only hashtag rules.

What This Means for Khoj

The existing khojai.zuhaib.pro project is closer to the desired product experience for khoj.zuhaib.pro than the current executor page. It already has the school front end, lessons, generated images, age groups, learning cycle, WhatsApp CTA, and agent pipeline.

Recommended direction: make khoj.zuhaib.pro the polished public product site by proxying/importing the KhojAI experience or migrating the KhojAI app into the main Khoj repo. Keep lab.zuhaib.pro/khoj as the internal progress dashboard.

Priority Fixes

Priority	Fix	Reason
P0	Decide proxy vs code migration for khoj.zuhaib.pro.	Main product direction decision.
P0	Bring khojai-aiteacher online only after safe log review/session confirmation.	Teacher bot is currently stopped.
P0	Add post-generation sanitizer and JSON validator.	Current output violates no-emoji/Urdu-only rules.
P1	Move hardcoded founder WhatsApp recipient into .env.	Safer and reusable.
P1	Retire or isolate pipeline.js if scheduler.js is source of truth.	Avoid overlapping generation.
P1	Fix security.js disk check and PM2 health logic.	Current checks can misreport health.
P2	Create clean Git sync for /var/www/khojai excluding secrets/db/logs/auth/generated output.	Stops local/VPS drift.

Secrets, private WhatsApp logs, token files, full database records, and Baileys auth contents were intentionally not opened or included.